



Stream Processing in Apache Kafka

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MLSECOPS IN ACTION COURSE



Building a Sophisticated Stream Topology

APIS OF KAFKA:

- Producer API
- Consumer API
- Share Consumer API
- Streams API
- Connect API
- Admin API



The **Share Consumer API** is Kafka's newer way to let multiple consumers read from the *same* partition at the same time.

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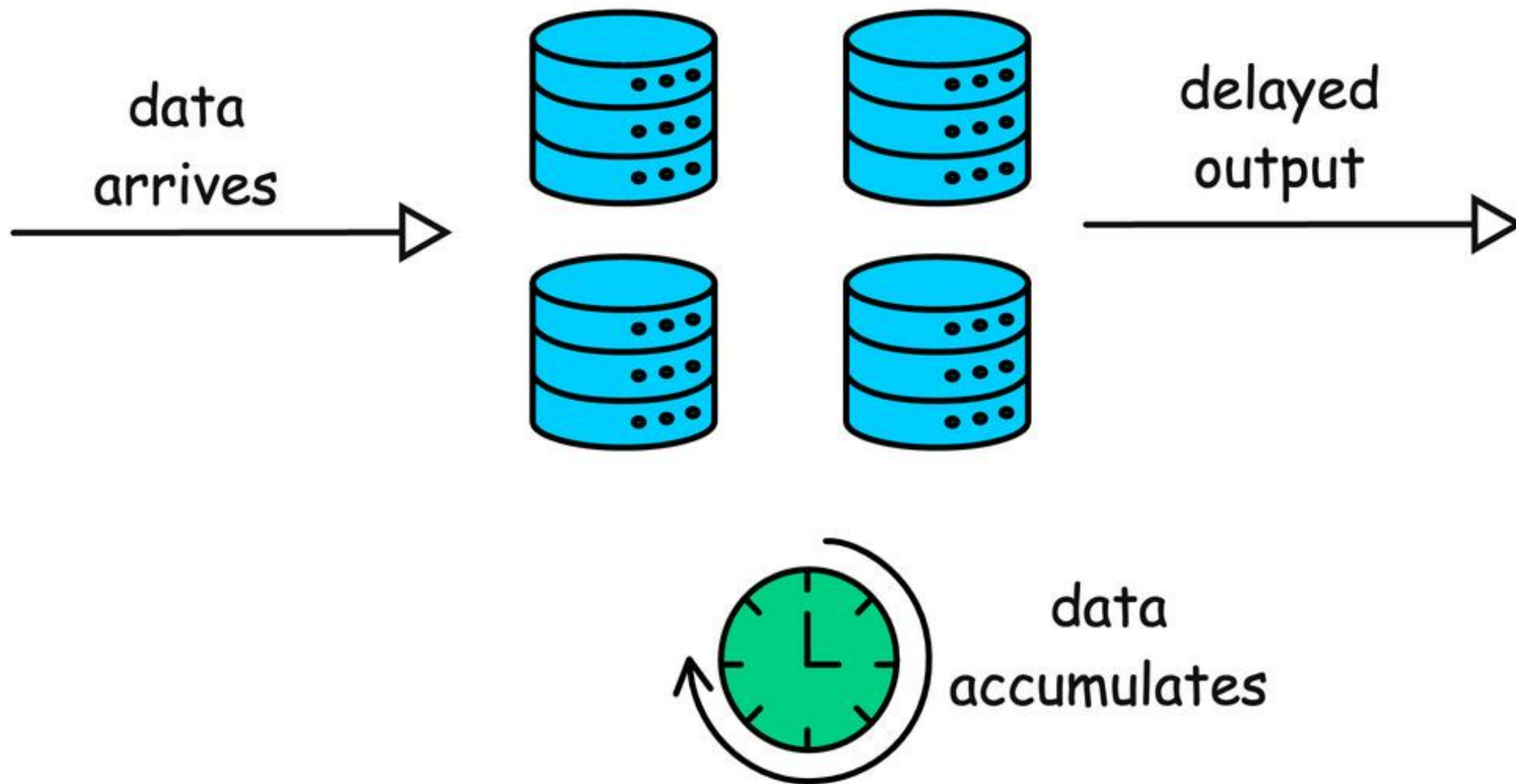


Quix
Streams

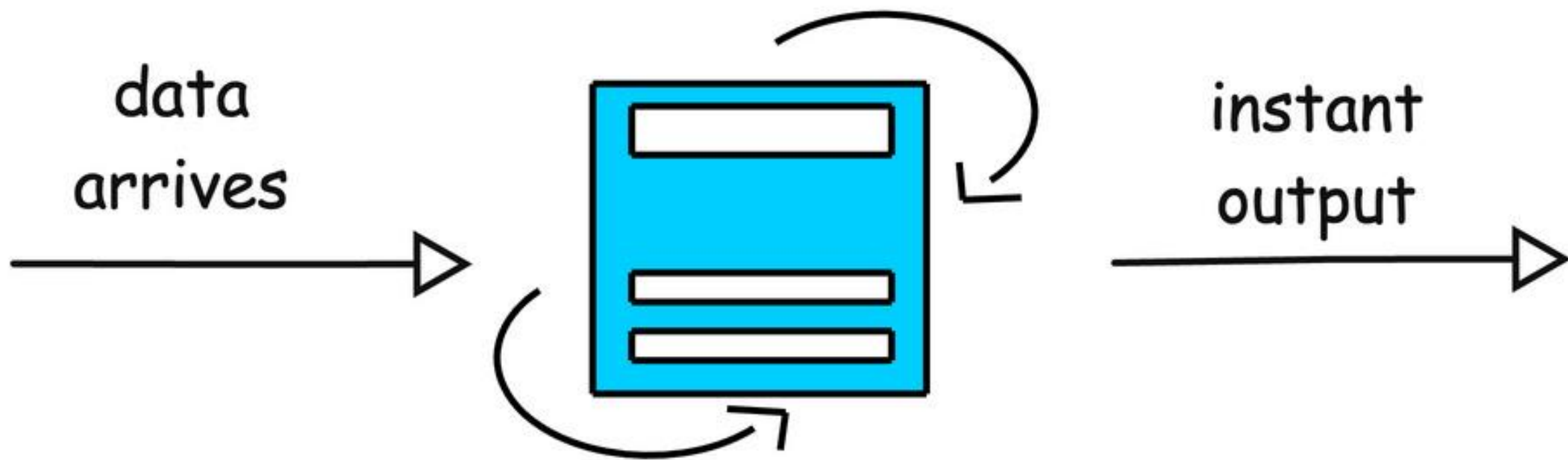
Before going further, we need to know some terms

What is Stream Processing?
And what is Batch Processing?

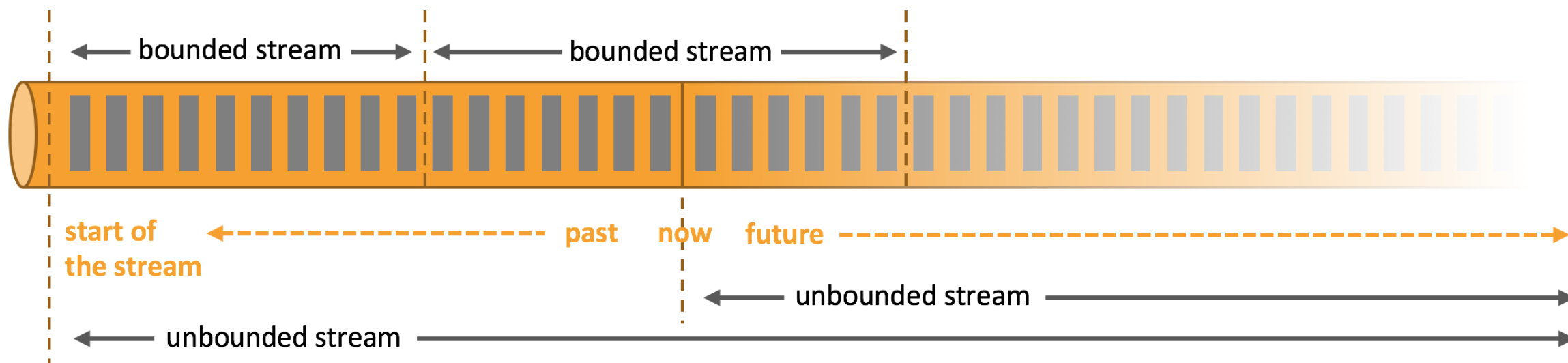
Batch Processing



Stream Processing



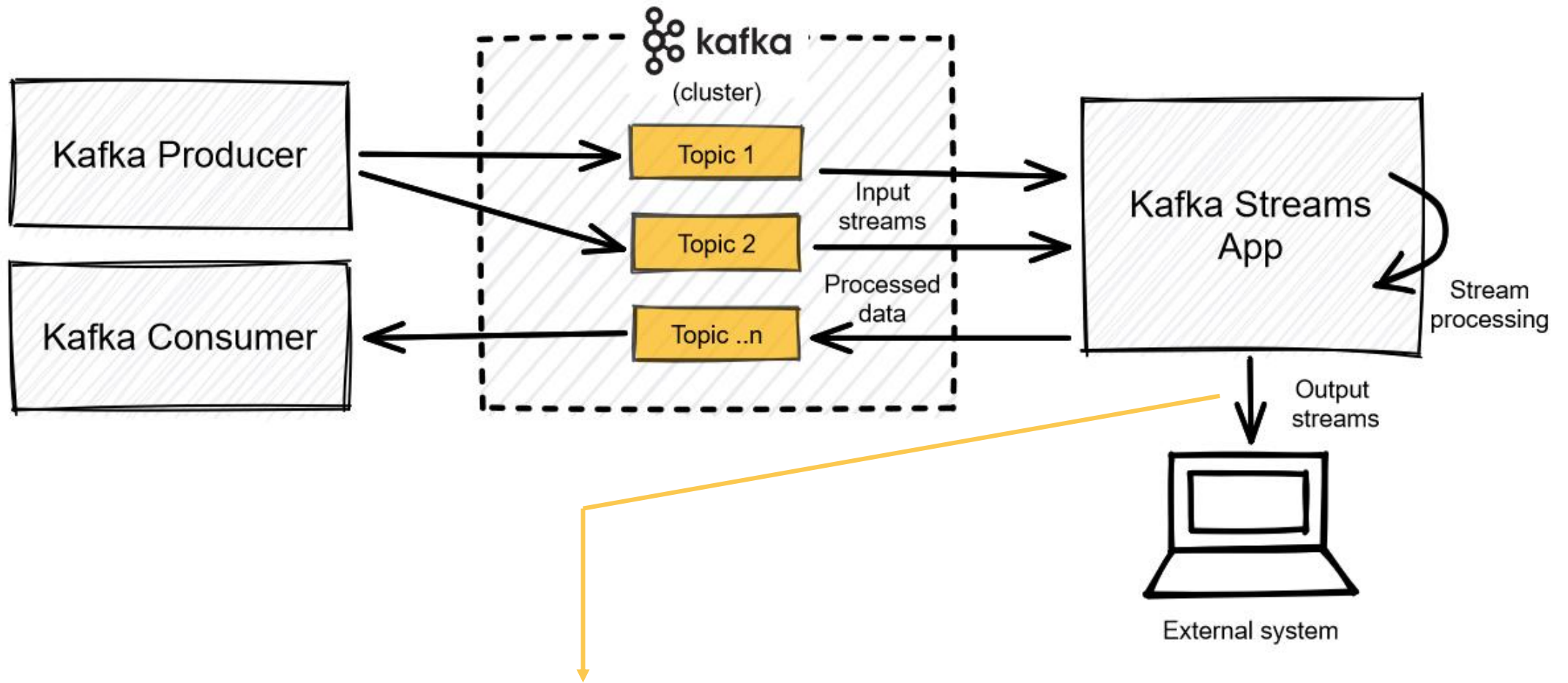
Stream processing (or streaming) is a process or application you implement that deals with an **uninterrupted flow of data** and performs work as soon as that data arrives.



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Kafka Streams = { **Streams DSL API (A high-level abstraction)**
Processor API (A low-Level API)

DSL comes from Domain-Specific Language



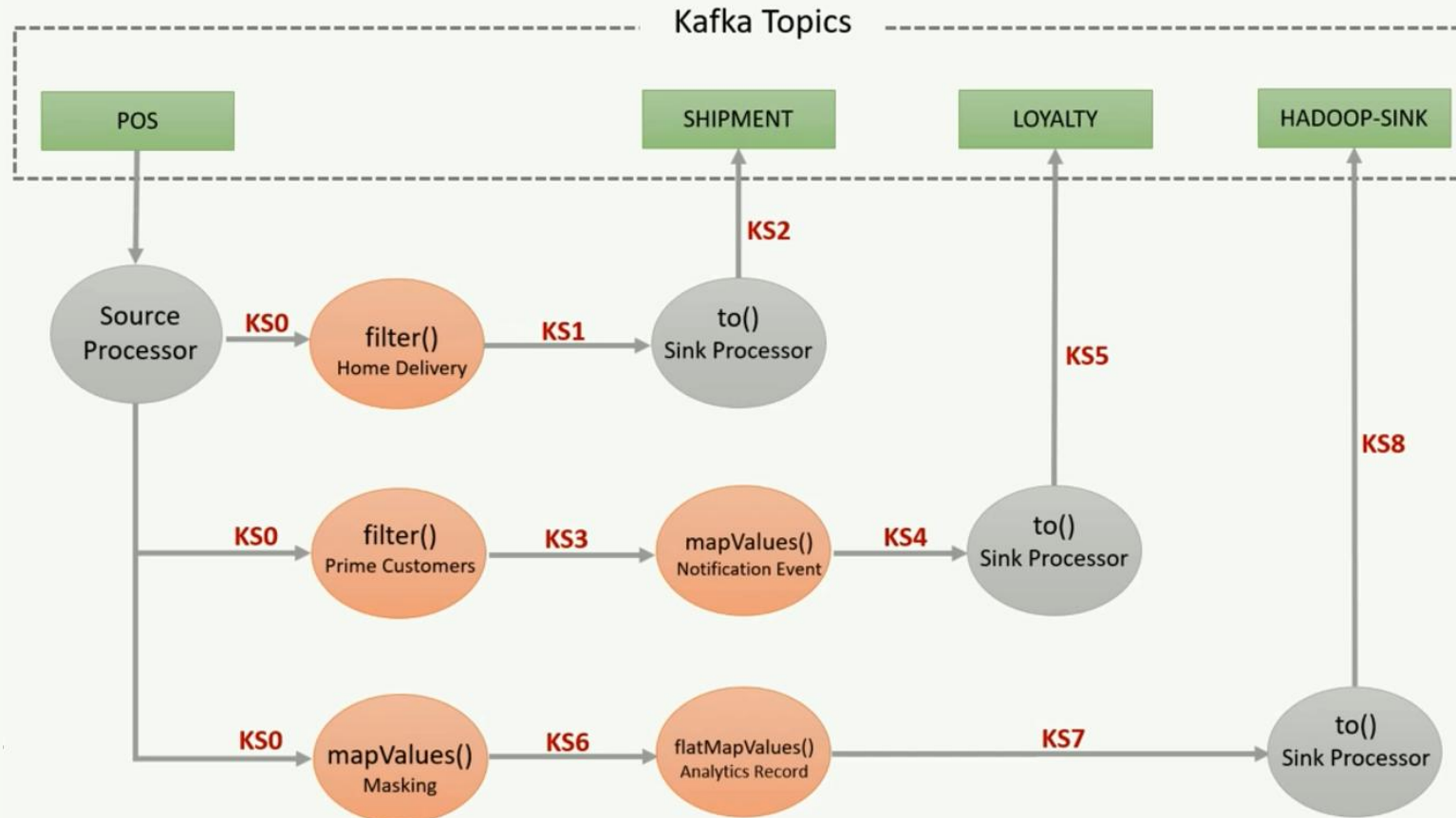
We can do this by using **Connect Sink Connectors**

What is Streams Topology?

Step-by-step computational logic of a stream processing application.

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Step-by-step computational logic of a stream processing application.



What is the STORY?

Business Requirement

XYZ is a Home Furniture and Kitchen utensils retailer. They have twenty retail stores spread all over the country. XYZ management decided to transform themselves into a real-time data-driven organization. As a first step towards that goal, they started sending their invoices to a Kafka cluster in real-time. The POS machines in all their stores are now sending invoices to a Kafka topic – POS. As a next step, they want to create the following automated services.

1. Shipment Service
2. Loyalty Management Service
3. Trend Analytics

Requirement Extract

1. Select Invoices where DeliveryType = "HOME-DELIVERY" and push them to the shipment service queue.
2. Select Invoices where CustomerType = "PRIME" and create a notification event for the Loyalty Management Service. The format for the new notification event is given here.
3. Select all Invoices, mask the personal information, and create records for Trend Analytics. When the records are ready, persist them to Hadoop storage for batch analytics. The format for the new Hadoop record is also given.

Kafka Topics

POS

Kafka Topics

POS

Source
Processor

```
graph TD; POS[POS] --> SP((Source Processor));
```

The diagram illustrates a data flow process. A green rectangular box labeled 'POS' is positioned at the top left. A solid gray arrow points vertically downwards from the 'POS' box to a gray circular node labeled 'Source Processor'. A large dashed rectangular box encloses the 'POS' box and extends horizontally to the right, with the text 'Kafka Topics' centered above its right side.

Kafka Topics

POS

Source
Processor

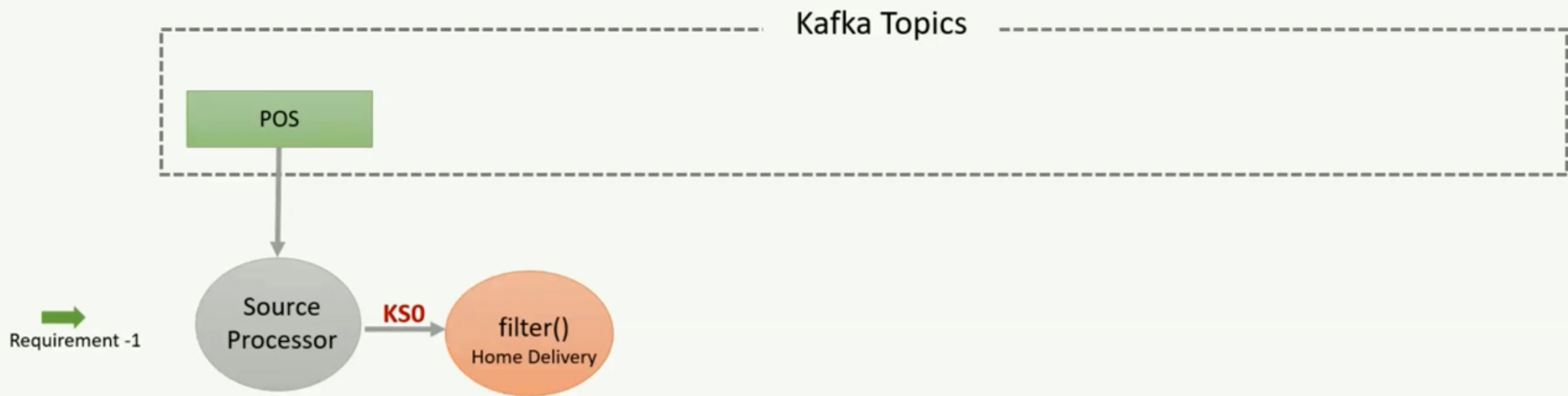
KSO

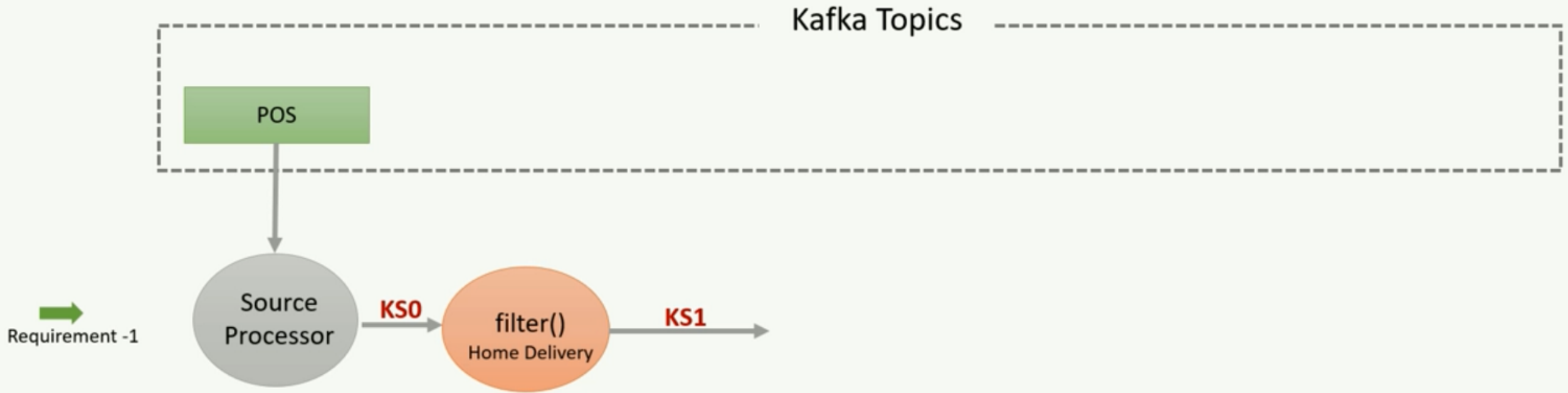
```
graph TD; subgraph Kafka_Topics [Kafka Topics]; POS[POS]; end; POS --> SP((Source Processor)); SP -- KSO --> Kafka_Topics;
```

The diagram illustrates a data processing pipeline. A green rectangular box labeled 'POS' is located inside a dashed-line container labeled 'Kafka Topics'. A vertical arrow points from the 'POS' box to a gray circular node labeled 'Source Processor'. From the 'Source Processor', a horizontal arrow points to the right, labeled with the text 'KSO' in red, indicating the output or connection to the Kafka Topics.

Requirement Extract

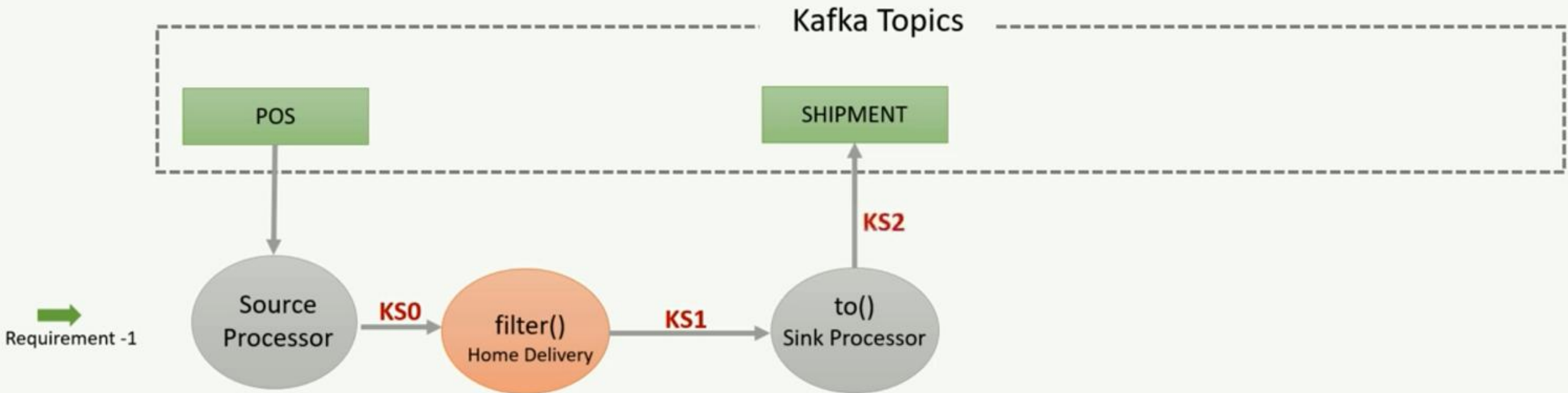
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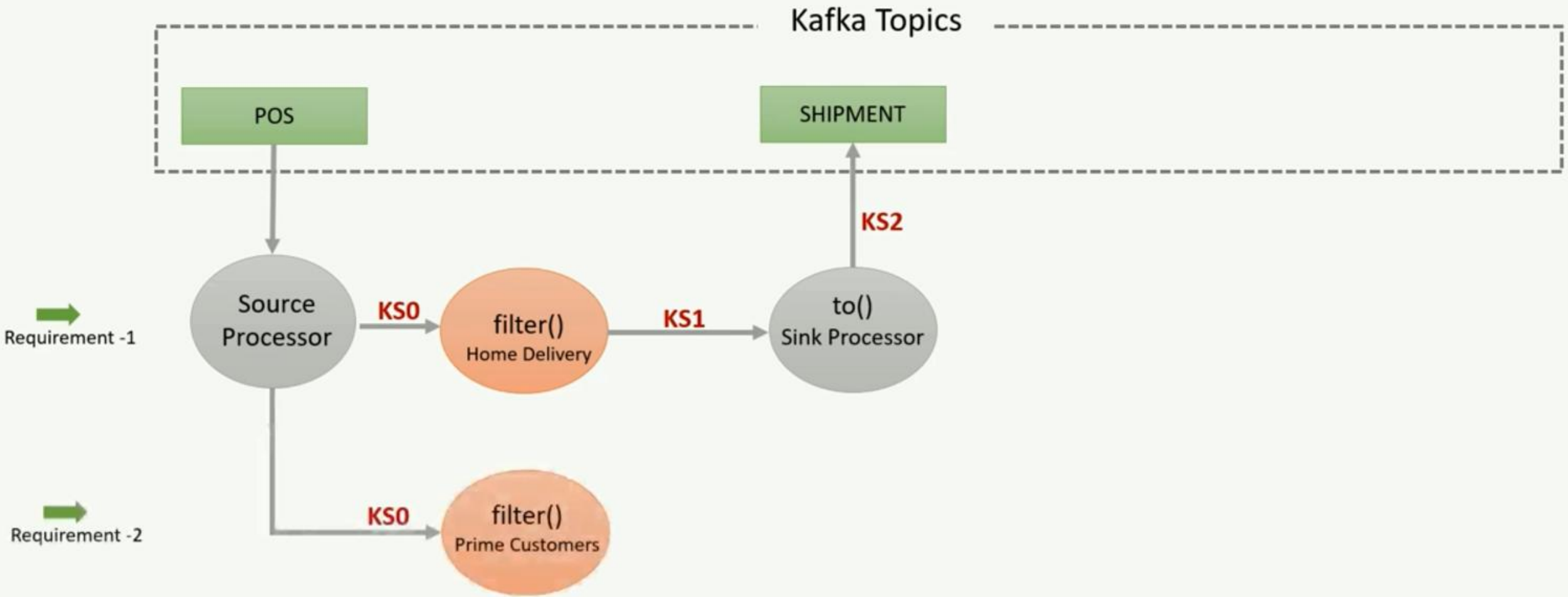
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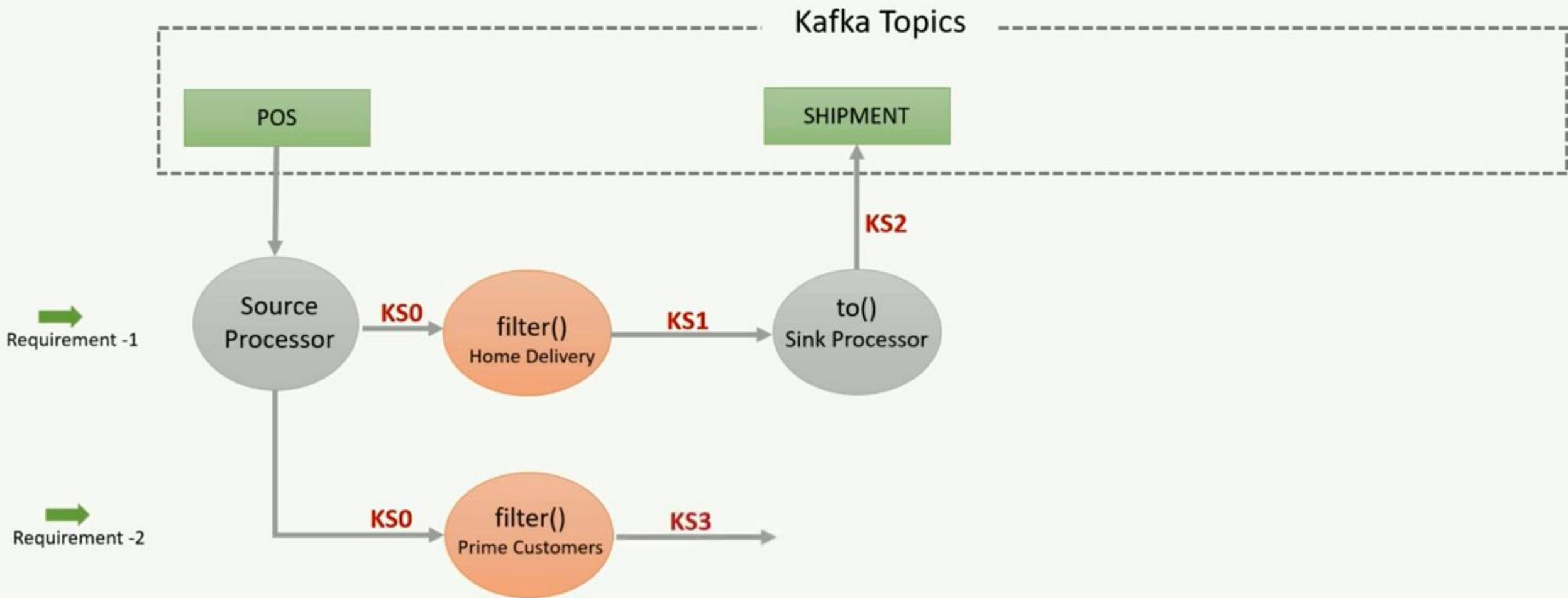
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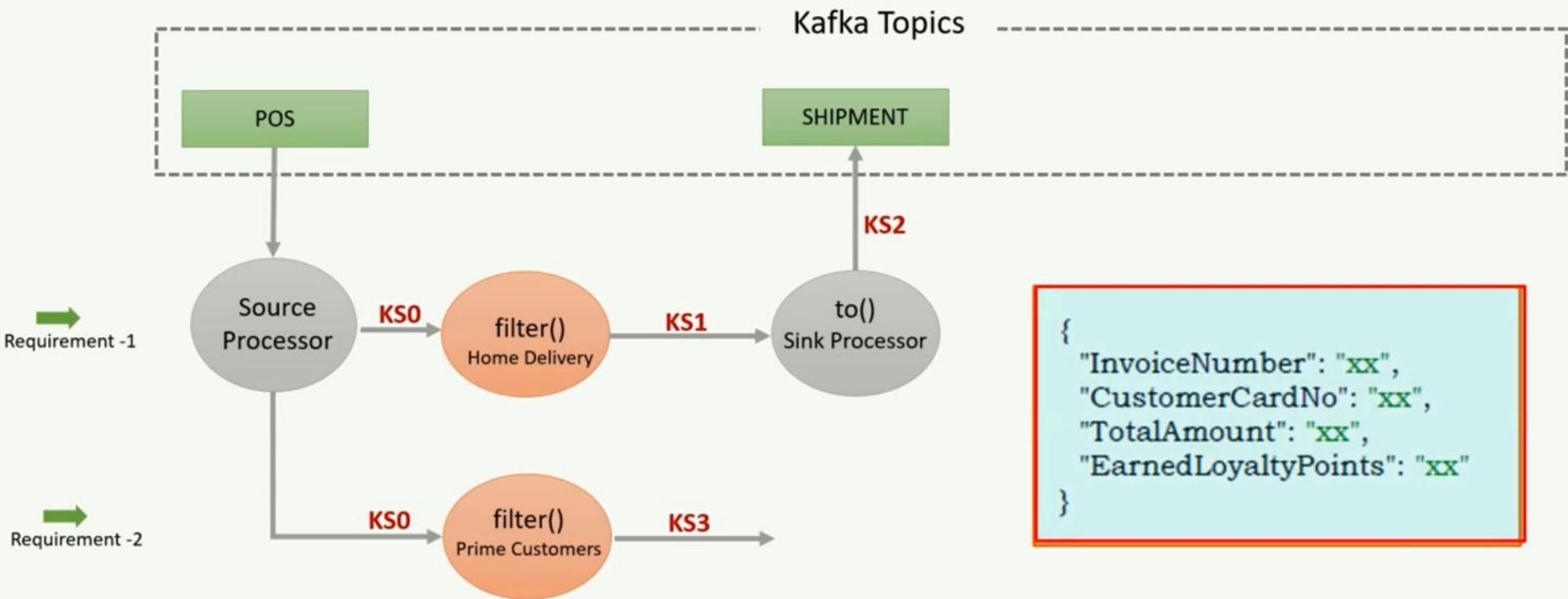


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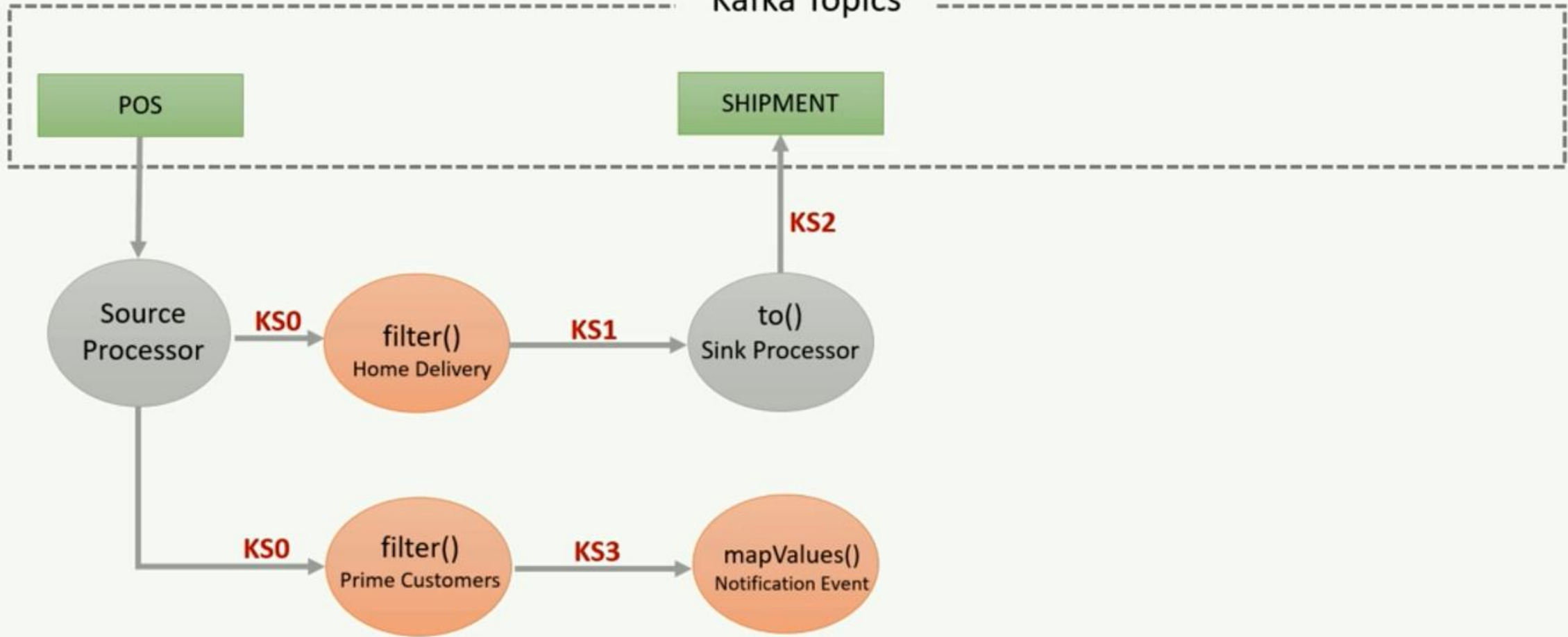
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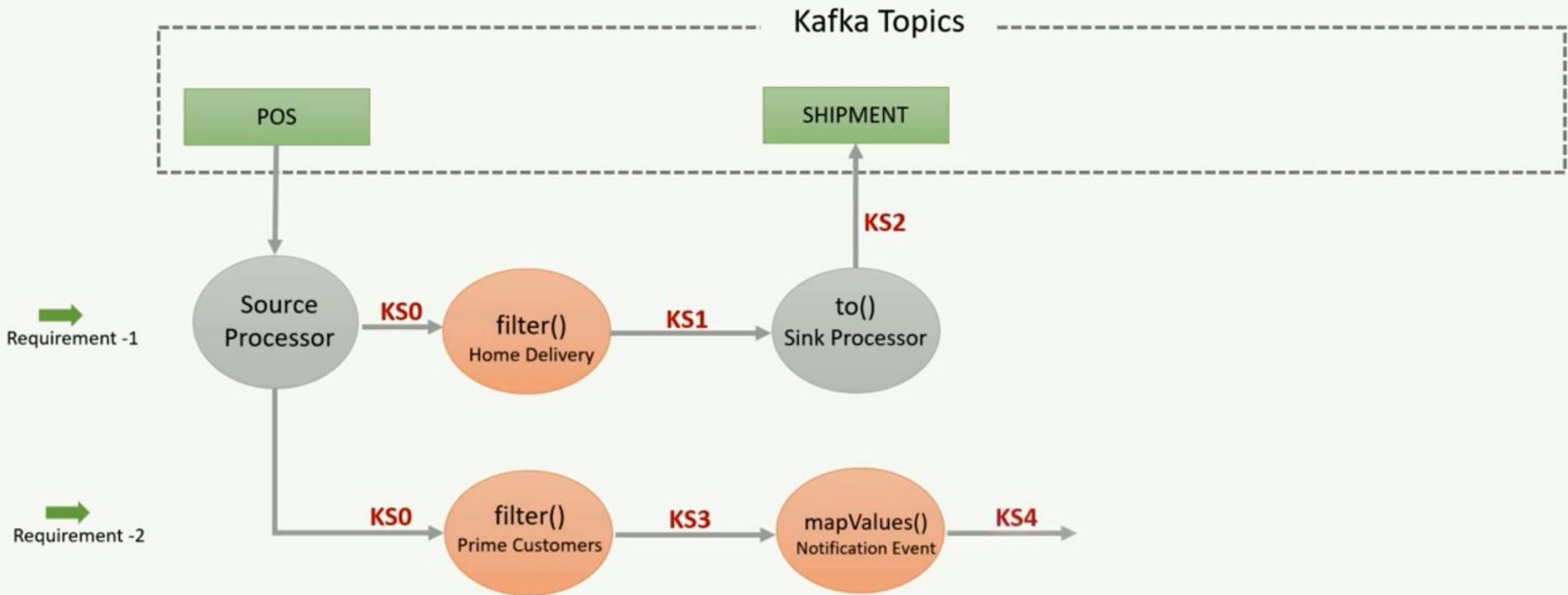






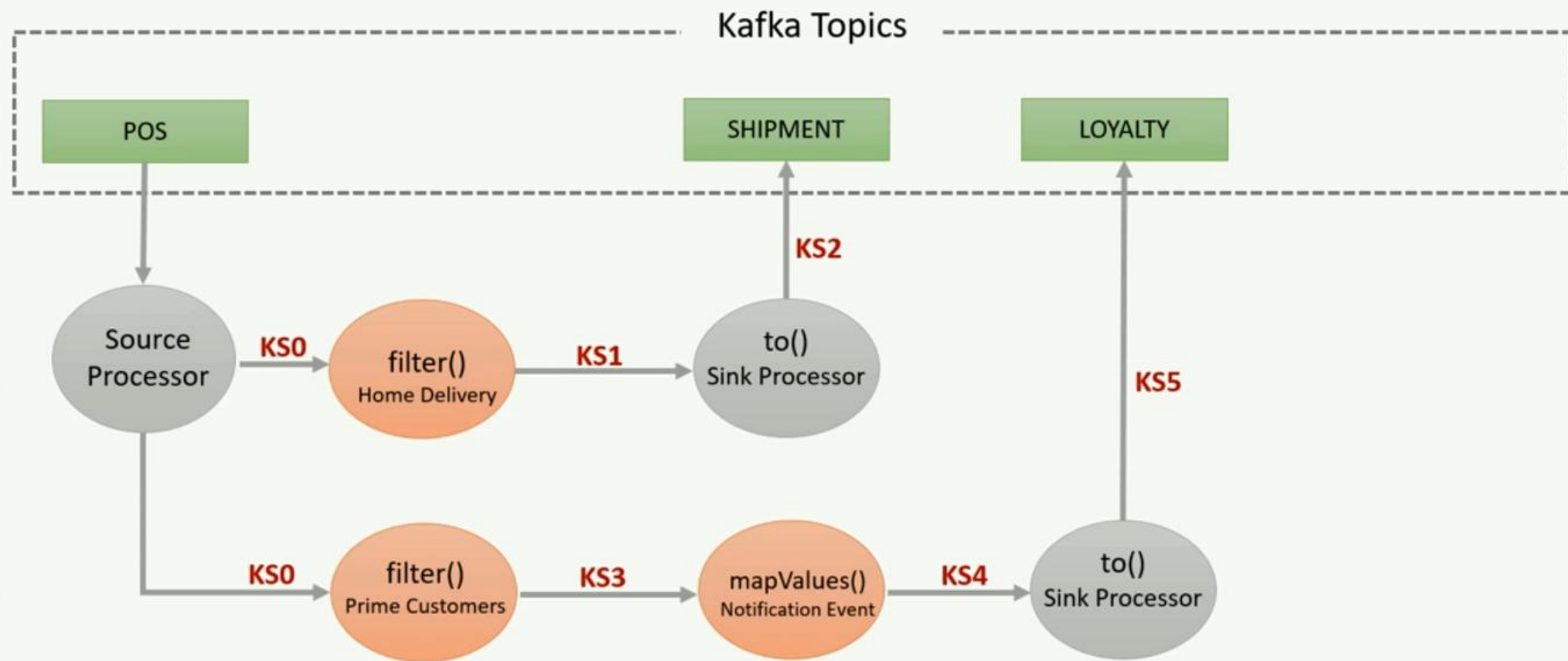
Kafka Topics





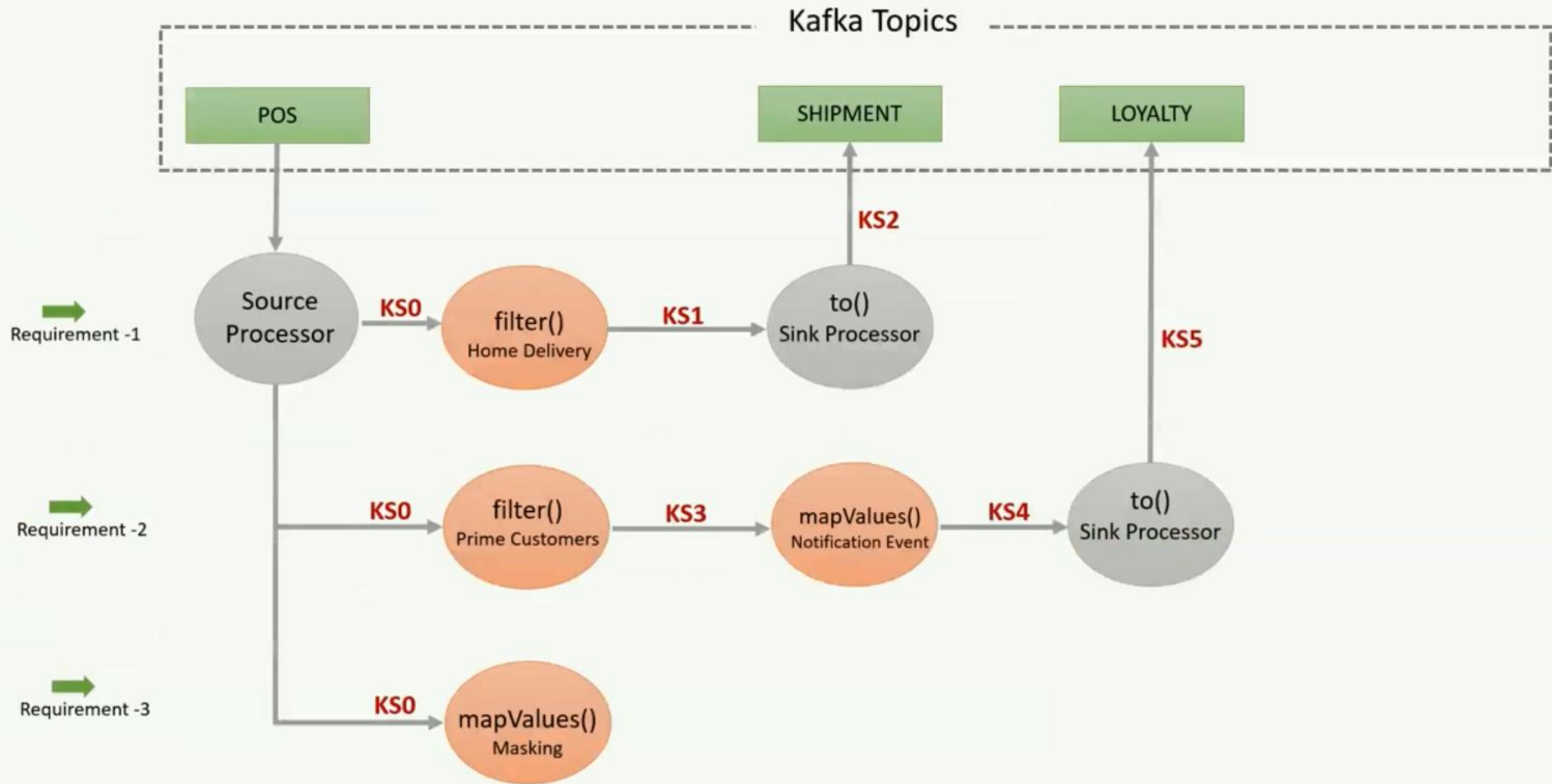
Requirement Extract

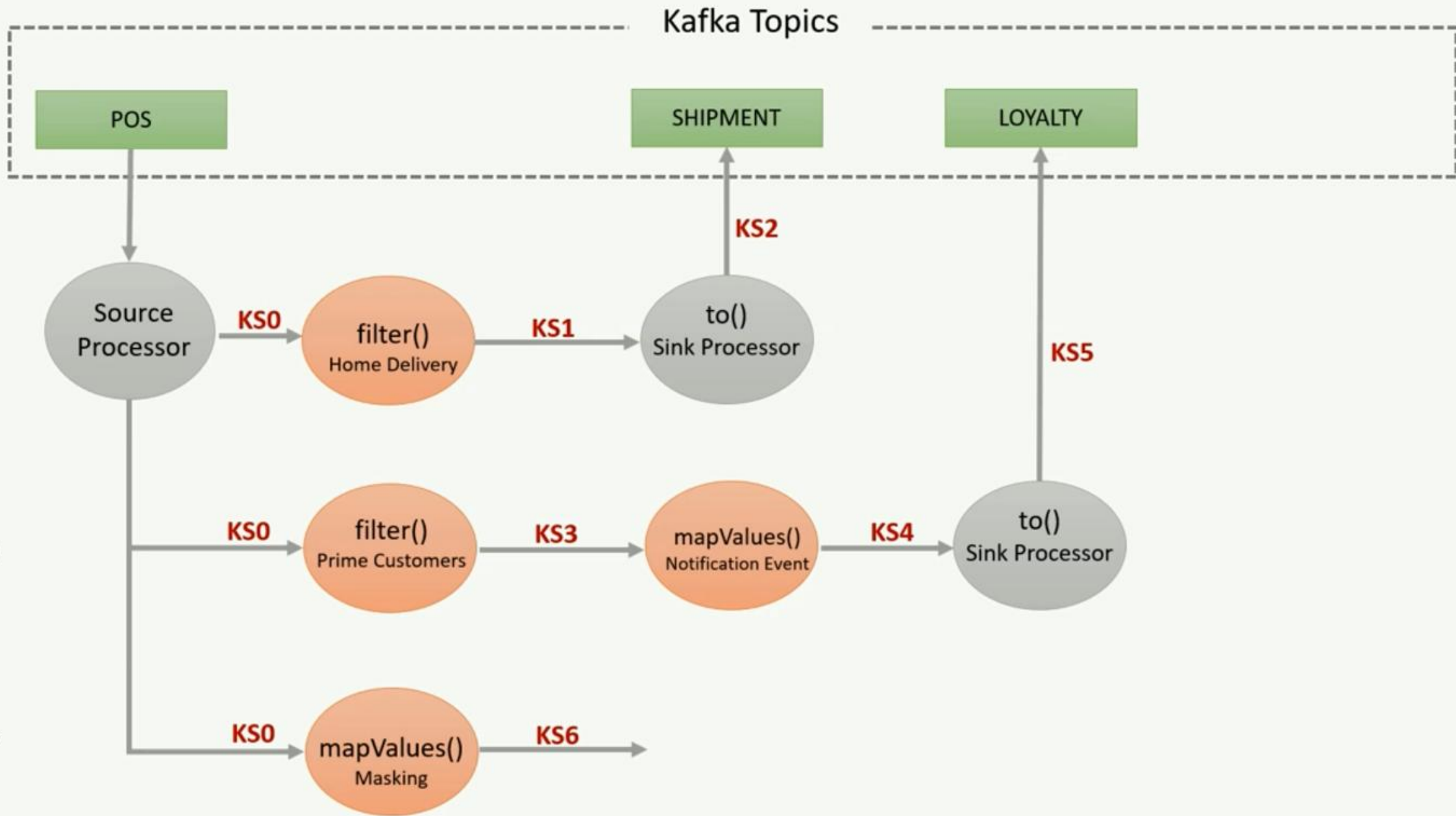
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```
{  
  "InvoiceNumber": "xx",  
  "CreatedTime": "xx",  
  "StoreID": "xx",  
  "PosID": "xx",  
  "CustomerType": "xx",  
  "PaymentMethod": "xx",  
  "DeliveryType": "xx",  
  "City": "xx",  
  "State": "xx",  
  "PinCode": "xx",  
  "ItemCode": "xx",  
  "ItemDescription": "xx",  
  "ItemPrice": "xx",  
  "ItemQty": "xx",  
  "TotalValue": "xx"  
}
```

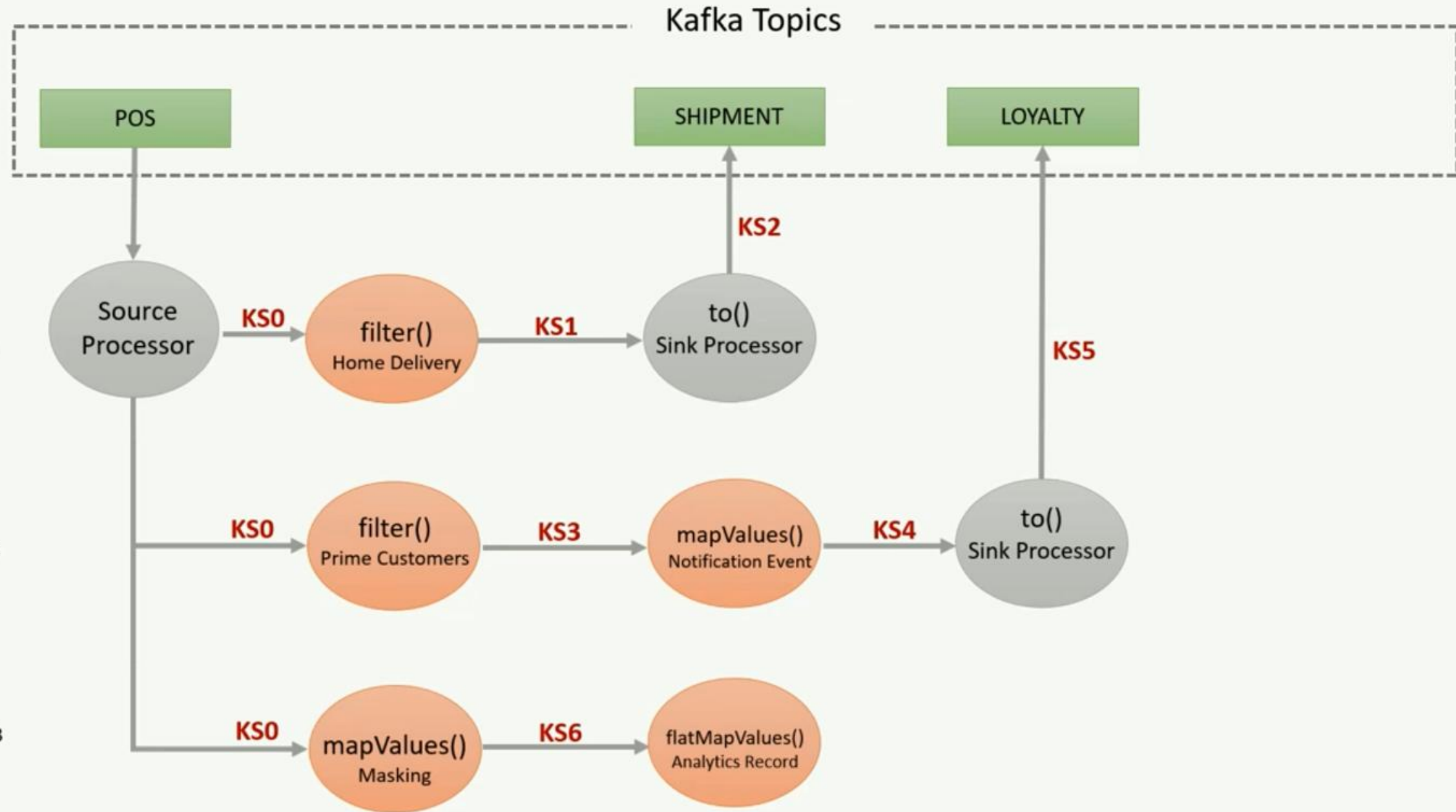
Sample Invoice

```
{  "InvoiceNumber": "2798495",
    "CreatedTime": 1552393659955,
    "StoreID": "STR7188",
    "PosID": "POS825",
    "CashierID": "OAS329",
    "CustomerType": "PRIME",
    "CustomerCardNo": "7051101351",
    "TotalAmount": 3529.0,
    "NumberOfItems": 2,
    "PaymentMethod": "CASH",
    "TaxableAmount": 3529.0,
    "CGST": 88.22500000000001,
    "SGST": 88.22500000000001,
    "CESS": 4.41125,
    "DeliveryType": "HOME-DELIVERY",
    "DeliveryAddress": {
        "AddressLine": "House No 727, 9696 Ullamcorper, Road",
        "City": "Dabgram",
        "State": "West Bengal",
        "PinCode": "953658",
        "ContactNumber": "4166559042"
    },
    "InvoiceLineItems": [
        {
            "ItemCode": "213",
            "ItemDescription": "Infant bed",
            "ItemPrice": 1755.0,
            "ItemQty": 1,
            "TotalValue": 1755.0
        },
        {
            "ItemCode": "628",
            "ItemDescription": "Window Scarf",
            "ItemPrice": 1774.0,
            "ItemQty": 1,
            "TotalValue": 1774.0
        }
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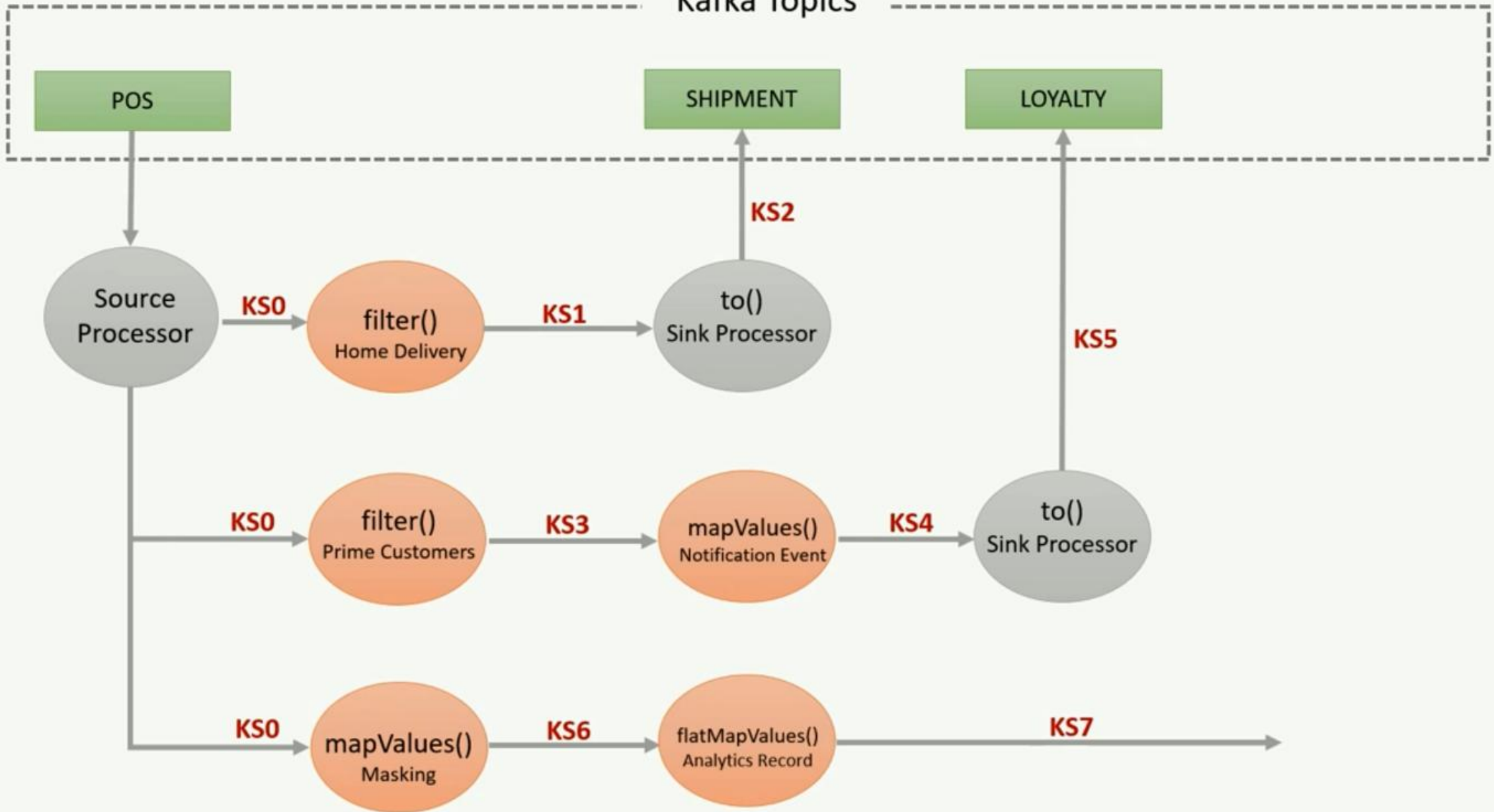

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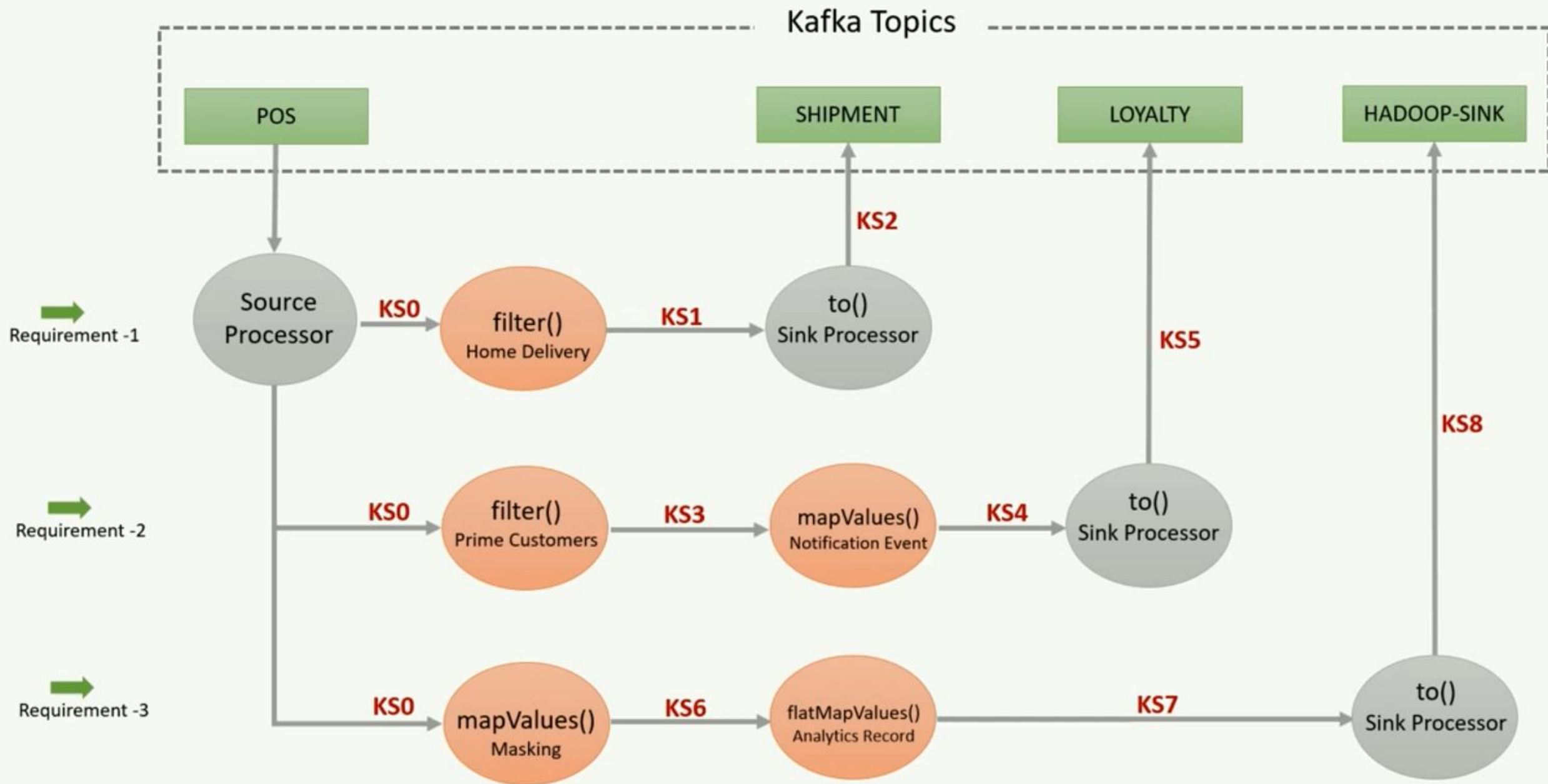


Kafka Topics



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Kafka is really useful in MLSecOps?

مثال‌های قبلی همه در دنیای تجارت الکترونیک
بودند، چه ربطی به امنیت دارند؟

Kafka is really useful in MLSecOps?

- 195.178.110.137 - - [01/Sep/2026:18:47:14 +0330] "GET /.env.backup HTTP/1.1" 301 795 "-" "fasthttp"
- 195.178.110.137 - - [01/Sep/2026:18:47:13 +0330] "GET /.env.old HTTP/1.1" 301 795 "-" "fasthttp"
- 195.178.110.137 - - [01/Sep/2026:18:47:13 +0330] "GET /.env.save HTTP/1.1" 301 795 "-" "fasthttp"
- 195.178.110.137 - - [01/Sep/2026:18:47:13 +0330] "GET /.env.dev HTTP/1.1" 301 795 "-" "fasthttp"
- 195.178.110.137 - - [01/Sep/2026:18:47:12 +0330] "GET /.env.local HTTP/1.1" 301 795 "-" "fasthttp"
- 195.178.110.137 - - [01/Sep/2026:18:47:12 +0330] "GET /.env.production HTTP/1.1" 301 795 "-" "fasthttp"
- 88.218.47.247 - - [01/Sep/2026:16:03:00 +0330] "POST /?rest_route=/batch/v1 HTTP/1.1" 301 795 "http://askmedot.ir/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36"

سرویس هوشمند یا API که ارائه دادیم هر لحظه توسط هزاران بات اسکن می‌شه (فاز شناسایی. به GET دقت کنید). ما می‌تونیم این پردازش لاگ سرور رو هم بصورت بچ انجام بدیم (مثلن روزی دوبار) و هم بصورت استریم و در لحظه رصد کنیم که اون API ما چقدر زیر فشار و درخواست هست. **اینجاست که وجود یک تکنولوژی مثل کافکا در استک واقعن عالیست. براحتی می‌تونیم لاگ‌ها رو هدایت کنیم سمت یک یا چند تاپیک کافکا و با quix اونها رو آنالیز کنیم.**

Coding Time. Fasten Your Seat Belt!